

3 (**short**) The speed, V , of a cylindrical water jet with cross-sectional area A and density ρ is to be measured by placing a cone with half-angle θ on the centreline of the jet and measuring the force, F , required to keep the cone stationary, as shown in Fig. 3. The jet can be considered to be inviscid.

- (a) Explain why the water can be assumed to leave the rear edge of the cone at speed V . [2]
- (b) With a simple model, calculate the horizontal force, F , on the cone. Sketch F as a function of θ for $0 < \theta < \pi$. When and why will this simple model break down? [4]
- (c) Downstream of the cone, the conical sheet of water has radius r and thickness t , where $r \gg t$. Neglecting gravitational effects, derive an expression for t as a function of r . [4]

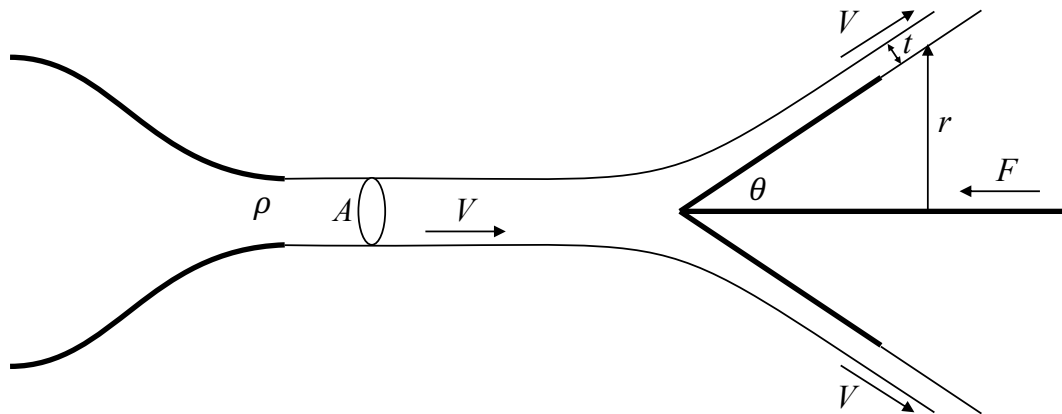


Fig. 3